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Non-linear single axis navigation sensor with strain relief

Abstract

The distal end of the catheter can be constructed to include one or more features to provide strain relief to wiring of multiple single axis sensors. In some examples, the multiple single axis sensors and associated wiring can be manufactured over a flexible tube that can be placed over a movable support member. In some examples, wiring can be wound an increased number of consecutive traverse turns on a distal and/or proximal side of a single axis sensor, and a shrink sleeve may be positioned over the traverse turns. In some examples a wire shield transition point can be positioned on a straight portion in a proximal direction to the distal portion of the catheter.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4815478	12/1988	Buchbinder et al.	N/A	N/A
RE34502	12/1993	Webster, Jr.	N/A	N/A
5275151	12/1993	Shockey et al.	N/A	N/A
5391199	12/1994	Ben-Haim	N/A	N/A
5395329	12/1994	Fleischhackor et al.	N/A	N/A
5443489	12/1994	Ben-Haim	N/A	N/A
5645065	12/1996	Shapiro et al.	N/A	N/A
5656029	12/1996	Imran et al.	N/A	N/A
5891088	12/1998	Thompson et al.	N/A	N/A
5997473	12/1998	Taniguchi et al.	N/A	N/A
6123699	12/1999	Webster, Jr.	N/A	N/A
6171277	12/2000	Ponzi	N/A	N/A
6183463	12/2000	Webster, Jr.	N/A	N/A
6484118	12/2001	Govari	N/A	N/A
6491681	12/2001	Kunis et al.	N/A	N/A
6500129	12/2001	Mahurkar	N/A	N/A
6554794	12/2002	Mueller et al.	N/A	N/A
6574492	12/2002	Ben-Haim et al.	N/A	N/A
6690963	12/2003	Ben-Haim et al.	N/A	N/A
6788967	12/2003	Ben-Haim et al.	N/A	N/A
6892090	12/2004	Verard et al.	N/A	N/A
6961602	12/2004	Fuimaono et al.	N/A	N/A
7174202	12/2006	Bladen et al.	N/A	N/A
7274957	12/2006	Drysen	N/A	N/A
7615044	12/2008	Scheibe et al.	N/A	N/A
7931616	12/2010	Selkee	N/A	N/A
8118775	12/2011	Grunewald et al.	N/A	N/A
8259505	12/2011	Oyama	N/A	N/A
8382662	12/2012	Soper et al.	N/A	N/A
8529505	12/2012	Grunewald et al.	N/A	N/A
8700133	12/2013	Hann	N/A	N/A
8792962	12/2013	Esguerra et al.	N/A	N/A
8805472	12/2013	Iglesias	N/A	N/A
8880147	12/2013	Tegg et al.	N/A	N/A

8926528	12/2014	Govari et al.	N/A	N/A
8936583	12/2014	Holzbauer et al.	N/A	N/A
10405774	12/2018	Esguerra Wilczynski et al.	N/A	N/A
2003/0028096	12/2002	Niwa et al.	N/A	N/A
2003/0160721	12/2002	Gilboa et al.	N/A	N/A
2005/0274425	12/2004	Ostrander et al.	N/A	N/A
2007/0164900	12/2006	Schneider et al.	N/A	N/A
2008/0255540	12/2007	Selkee	N/A	N/A
2009/0005754	12/2008	Soetermans	N/A	N/A
2009/0192412	12/2008	Sela et al.	N/A	N/A
2010/0036285	12/2009	Govari	600/377	A61B 5/062
2010/0069834	12/2009	Schultz	N/A	N/A
2010/0168827	12/2009	Schultz	N/A	N/A
2010/0222664	12/2009	Lemon et al.	N/A	N/A
2011/0054287	12/2010	Schultz	N/A	N/A
2012/0143088	12/2011	Schultz	N/A	N/A
2012/0172842	12/2011	Sela et al.	N/A	N/A
2014/0228838	12/2013	Kirschenman	N/A	N/A
2014/0275957	12/2013	Lupotti	N/A	N/A
2020/0015703	12/2019	Esguerra Wilczynski	N/A	A61B 5/341
2021/0001084	12/2020	Highsmith et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1 457 226	12/2003	EP	N/A
1 562 665	12/2003	EP	N/A
2 151 209	12/2008	EP	N/A
2 165 730	12/2008	EP	N/A
2 301 617	12/2009	EP	N/A
2 460 558	12/2010	EP	N/A
2 897 524	12/2013	EP	N/A
2010-36040	12/2009	JP	N/A
WO 96/34652	12/1995	WO	N/A
WO 2007/130720	12/2006	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report dated Sep. 28, 2022, from corresponding European Application No. 22170560.1. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority under the Paris Convention as well as 35 U.S.C. §§ 119 and 120 to prior filed U.S. Provisional Patent Application No. 63/181,641 filed on Apr. 29, 2021 which is hereby incorporated by reference as set forth in full herein.

FIELD

(1) This invention relates to a catheter, in particular, a catheter having location sensors mounted on a flexible, non-linear distal end portion for improved position sensing of the distal end portion.

BACKGROUND

(2) A magnetic field can be sensed by positioning a conductive coil in the magnetic field and observing electrical current and/or voltage induced in the coil by a change in the magnetic field that is aligned with an axis of the conductive coil. Because electrical current is induced in the conductive coil, such a coil is also referred to as an inductive coil. Relative position of a sensor including one or more inductive coils can be determined in relation to a known magnetic field source by monitoring the induced electric current and/or voltage of the inductive coil(s).

(3) An inductive coil wrapped around a flexible catheter structure can deflect causing strain on coil wiring. In such cases, kinking or breakage of wiring can occur.

SUMMARY

(4) Examples presented herein generally include a sensor assembly having inductive coils wound about a tubular member that can be positioned over a support member at a distal portion of a catheter. The sensor assembly can be part of a navigation system which uses electrical currents induced in the inductive coils in the sensor assembly to determine a position of the distal portion of the catheter when the distal portion is positioned within a known magnetic field. The tubular member can be configured to relieve strain on the inductive coils and/or wiring of the sensor assembly as the support member flexes to reshape the distal portion of the catheter. Each of the inductive coils can be co-axial to the tubular member, such that each inductive coil functions respectively as a single axis sensor (SAS).

(5) An example SAS assembly can include a tubular member, a distal inductive coil, a first dual cable, and a proximal inductive coil. The tubular member can have a lumen therethrough sized to receive an elongated support member suitable for shaping a distal section of a mapping catheter. The lumen can have a diameter measuring from approximately 0.1 mm to about 0.25 mm. The distal inductive coil can be affixed to the tubular member and coaxial to the tubular member. The distal inductive coil can include leads. The first dual cable can be electrically joined to the leads of the distal inductive coil and wound over the tubular member. The proximal inductive coil can be affixed to the tubular member across (over and/or under) the first dual cable. The proximal inductive coil can be coaxial to the tubular member. The proximal inductive coil can be separated from the distal inductive coil such that the distal inductive coil moves in relation to the proximal inductive coil when the tubular member is moved from a linear configuration to a nonlinear configuration. The nonlinear configuration can be approximately circular with a circumference measuring from approximately 30 millimeters to approximately 80 millimeters.

(6) The SAS assembly can further include an intermediate inductive coil affixed to the tubular member, coaxial to the tubular member, positioned between the distal inductive coil and the proximal inductive coil when the SAS assembly is in the linear configuration, and positioned such that the proximal inductive coil, intermediate inductive coil, and distal inductive coil collectively function as a three-axis sensor when the SAS assembly is in the nonlinear configuration.

(7) The SAS assembly can further include a second dual cable electrically joined to leads of the intermediate inductive coil and wound over the tubular member to cross (over and/or under) the proximal inductive coil.

(8) The nonlinear configuration can be approximately circular. The proximal inductive coil, intermediate inductive coil, and distal inductive coil can be positioned approximately equidistant

around a circumference of the nonlinear configuration.

(9) An example mapping catheter can include an elongated support member extending through a distal section of the mapping catheter and a SAS assembly. The SAS assembly of the example catheter can be configured similarly to the above example SAS assembly. The SAS assembly can be manufactured separately from the elongated support member, and the elongated support member can be inserted into the SAS assembly.

(10) The SAS assembly can include a tubular member surrounding the elongated support member, a distal inductive coil affixed to the tubular member, and a proximal inductive coil affixed to the tubular member. The elongated support member can have a nonlinear predetermined configuration to which the elongated support member moves when the distal section is disposed within a patient. The distal inductive coil and the proximal inductive coil can each be coaxial to the tubular member. The proximal inductive coil and the distal inductive coil can be separated from each other such that the distal inductive coil moves in relation to the proximal inductive coil when the elongated support member moves to the nonlinear predetermined configuration. The elongated support member can include a memory shape material heat set to the nonlinear predetermined configuration. The SAS sensor assembly can further include a first dual cable electrically joined to leads of the distal inductive coil and wound over the tubular member to cross (over and/or under) the proximal inductive coil.

(11) The SAS sensor assembly can further include intermediate inductive coil affixed to the tubular member, coaxial to the tubular member, and positioned between the distal inductive coil and the proximal inductive coil when the elongated support member is linearly elongated. The proximal inductive coil, intermediate inductive coil, and distal inductive coil can collectively function as a three-axis sensor when the elongated support member moves to the nonlinear predetermined configuration.

(12) The nonlinear predetermined configuration can be approximately circular. The proximal inductive coil, intermediate inductive coil, and distal inductive coil can be positioned approximately equidistant around a circumference of the nonlinear predetermined configuration. The nonlinear predetermined configuration can have a circumference measuring from approximately 10 millimeters to approximately 50 millimeters.

(13) The SAS sensor assembly can further include a second dual cable electrically joined to leads of the intermediate inductive coil and wound over the tubular member to cross the proximal inductive coil.

(14) Another example mapping catheter can include an elongated body, a distal section distal to the elongated body, an elongated support member extending through the distal section, a tubular member surrounding the elongated support member, a distal inductive coil affixed to the tubular member, and a proximal inductive coil affixed to the tubular member. The distal section can be movable from a linear delivery configuration to a nonlinear deployed configuration. The elongated support member can be shaped to move the distal section into the nonlinear deployed configuration. The elongated support member can include a memory shape material heat set to a nonlinear predetermined configuration that approximates the nonlinear deployed configuration of the distal section. The distal inductive coil and proximal inductive coil can respectively be coaxial to the tubular member. The proximal inductive coil can be separated from the distal inductive coil such that the distal inductive coil moves in relation to the proximal inductive coil when the distal section moves from the linear delivery configuration to the nonlinear deployed configuration. The mapping catheter can further include a first dual cable electrically joined to leads of the distal inductive coil and wound over the tubular member to cross (over and/or under) the proximal inductive coil.

(15) The mapping catheter can further include an intermediate inductive coil affixed to the tubular member, coaxial to the tubular member, and positioned between the distal inductive coil and the proximal inductive coil when the distal section is in the linear delivery configuration. The proximal

inductive coil, intermediate inductive coil, and distal inductive coil can collectively function as a three-axis sensor when the distal section is in the nonlinear deployed configuration.

(16) The nonlinear deployed configuration can be approximately circular. The proximal inductive coil, intermediate inductive coil, and distal inductive coil can be positioned approximately equidistant around a circumference of the distal section in the nonlinear deployed configuration. The nonlinear deployed configuration can have a circumference measuring from approximately 10 millimeters to approximately 50 millimeters.

(17) The mapping catheter can further include a second dual cable electrically joined to leads of the intermediate inductive coil and wound over the tubular member to cross the proximal inductive coil.

(18) The mapping catheter can further include a control handle proximal of the elongated body, a contraction wire, and/or ring electrodes circumscribing the distal section. The contraction wire can extend through the elongated body and distal section. The contraction wire can be manipulated to shape the nonlinear deployed configuration. The ring electrodes can be configured to receive electrical signals from tissue during a patient treatment.

(19) An example method can include some or all of the following steps that can be executed in various orders, and the method can include additional steps not listed. A first inductive coil can be wound around a tubular member positioned over a mandrel. A first dual cable electrically joined to leads of the first inductive coil can be wound around the tubular member. A second inductive coil can be wound around the tubular member such that the first dual cable crosses (over and/or under) the second inductive coil and the second inductive coil is separated from the first inductive coil such that when the tubular member is flexed into a nonlinear configuration the first inductive coil moves in relation to the second inductive coil.

(20) The method can further include flexing the tubular member, with the first inductive coil, first dual cable, and second inductive coil thereon into an approximately circular shape with a circumference measuring from approximately 30 millimeters to approximately 80 millimeters.

(21) The method can further include extending an elongated support member having a predetermined shape through a lumen of the tubular member. The elongated support member can be extended through the lumen of the tubular member after winding the first inductive coil around the tubular member, after winding the first dual cable around the tubular member, and after winding the second inductive coil around the tubular member. The method can further include moving the elongated support member into the predetermined shape thereby forcing the tubular member into the nonlinear configuration.

(22) The method can further include winding an intermediate inductive coil around the tubular member at a position between the first inductive coil and the second inductive coil when the tubular member is linear, such that the first inductive coil, intermediate inductive coil, and second inductive coil collectively function as a three-axis sensor when the tubular member is in the nonlinear configuration, and such that the first dual cable crosses (over and/or under) the intermediate inductive coil.

(23) The method can further include winding a second dual cable electrically joined to leads of the intermediate inductive coil around the tubular member to cross the second inductive coil.

(24) The method can further include positioning the first inductive coil, intermediate inductive coil, and second inductive coil such that when the tubular member is in a circular shape, the first inductive coil, intermediate inductive coil, and second inductive coil are approximately equidistant from each other around the circular shape.

(25) Another example mapping catheter can include coils of single axis sensors, dual cables electrically connected to a respective coil of a respective single axis sensor, and a shrink sleeve. Each dual cable can be respectively wound for about 5 to about 7 consecutive traverse turns of approximately 720° around a tubular member or directly around a support member on a distal and/or proximal side of the respective coil. The shrink sleeve can be positioned to completely cover

the traverse turns. The example mapping catheter can further include a shield wire transition point positioned on a straight region of a distal end section of the mapping catheter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other features and advantages of the present invention will be better understood by reference to the following detailed description when considered in conjunction with the accompanying drawings. It is understood that selected structures and features have not been shown in certain drawings so as to provide better viewing of the remaining structures and features.
- (2) FIG. 1 is a top plan view illustration of an example nonlinear single axis sensor assembly in accordance with the present invention.
- (3) FIG. 2 is a side view illustration of the assembly in FIG. 1 with certain components removed for the sake of illustration.
- (4) FIG. 3 is a side cut-away view illustration of an example distal sensor of the assembly in FIG. 1.
- (5) FIG. 4 is a side cut-away view illustration of an example of an intermediate sensor of the assembly in FIG. 1.
- (6) FIG. 5 is a side cut-away view illustration of an example proximal sensor of the assembly in FIG. 1.
- (7) FIG. 6 is a top plan view illustration an example catheter in accordance with the present invention.
- (8) FIG. 7 is a side view illustration of a distal portion of the catheter of FIG. 6, including an intermediate section and a mapping assembly.
- (9) FIG. 8 is a longitudinal cross-sectional view illustration of the mapping assembly of FIG. 7, taken along line 8-8.
- (10) FIG. 9 is a longitudinal cross-sectional view illustration of the intermediate section of FIG. 7, taken along line 9-9.
- (11) FIG. 10 is a side cross-sectional view illustration of an example distal end of the catheter of FIG. 6.
- (12) FIG. 11 is a side cross-sectional view illustration of an example junction of a catheter body and an intermediate section, taken along a first diameter.
- (13) FIG. 12 is a side cross-sectional view illustration of the junction of FIG. 11, taken along a second diameter generally perpendicular to the first diameter.
- (14) FIGS. 13A through 13E is a flow diagram illustrating steps of constructing a distal end of an example catheter in accordance with the present invention.
- (15) FIG. 14 is a side view illustration of an intermediate sensor of a variation of the example catheter in accordance with the present invention.
- (16) FIG. 15 is a side view illustration of an intermediate sensor of another variation of the example catheter in accordance with the present invention.
- (17) FIG. 16 is a side view illustration of a variation of the example nonlinear single axis sensor assembly shown in FIG. 2 with certain components removed for the sake of illustration in accordance with the present invention.

DETAILED DESCRIPTION

- (18) As used herein, the terms “about” or “approximately” for any numerical values or ranges indicate a suitable dimensional tolerance that allows the part or collection of components to function for its intended purpose as described herein. More specifically, “about” or “approximately” may refer to the range of values $\pm 20\%$ of the recited value, e.g. “about 90%” may refer to the range of values from 71% to 99%.

(19) As used herein, the terms “tubular” and “tube” are to be construed broadly and are not limited to a structure that is a right cylinder or strictly circumferential in cross-section or of a uniform cross-section throughout its length. For example, the tubular structure or system is generally illustrated as a substantially right cylindrical structure. However, the tubular system may have a tapered or curved outer surface without departing from the scope of the present invention.

(20) FIG. 1 illustrates a top plan view of a partially assembled circular region **39** of a distal end section **15** of a catheter (see example catheter **10** illustrated in FIG. 6). The circular region **39** is illustrated as including a nonlinear single axis sensor (SAS) assembly **400** including three sensors **401A**, **401B**, **401C** configured to sense location and/or position of the circular region **39**, cables **409A**, **409B**, **409C** configured to carry electrical signals to and/or from each of the inductive sensors **401A**, **401B**, **401C** respectively, and a tubular member **404** upon which the sensors **401A**, **401B**, **401C** and the cables **409A**, **409B**, **409C** are mounted. A support member **54** can extend through the tubular member **404** providing structure and shape to the distal end section **15**. In some examples, the sensors **401A**, **401B**, **401C** and cables **409A**, **409B**, **409C** can be manufactured onto the tubular member **404** such the SAS assembly **400** is a component separate from the support member **54**. The SAS assembly **400** can then slide onto the support member **54**. The tubular member **404** can provide strain relief to the sensors **401A**, **401B**, **401C** and/or cables **409A**, **409B**, **409C** when the distal end section **15** flexes.

(21) Three single axis sensors **401A**, **401B**, **401C** are illustrated positioned at equidistance from each other along the generally circular main region **39**. A proximal sensor **401C** is distal of an elbow **37** where the distal end section **15** bends to join a catheter shaft **14**. An intermediate sensor **401B** is about 120 degrees from the proximal sensor **401C**. A distal sensor **401A** is about 120 degrees from the intermediate sensor. Configured as such, the three sensors **401A**, **401B**, **401C** can collectively function as a three-axis sensor. Details of each sensor **401A**, **401B**, **401C** are illustrated in FIGS. 3 through 5 respectively.

(22) FIG. 2 illustrates a side view of the distal end section **15** viewing the proximal sensor **401C** and proximal portions of cables **409A**, **409B**, **409C** with sensors and cable portions distal the proximal sensor **401C** omitted for the sake of illustration. The sensors **401A**, **401B**, **401C** enable the circular region **39** carrying the nonlinear SAS assembly **400** to be viewed under mapping systems such as those manufactured and sold by Biosense Webster, Inc., including the CARTO, CARTO XP, and NOGA mapping systems. Electrically, the sensors **401A**, **401B**, **401C** can function similarly or identically to single axis sensors described in U.S. Pat. Nos. 8,792,962, 10,405,774, and U.S. Patent Publication No. 2020/0015703 each incorporated by reference herein; U.S. Pat. No. 8,792,962 is attached in the Appendix of U.S. Provisional Patent Application No. 63/181,641 to which the present application claims priority.

(23) Referring collectively to FIGS. 1 and 2, the SAS assembly **400** can be configured to elongate to a substantially linear configuration when the distal end section **15** of the catheter is delivered through bodily lumens (e.g. blood vessels) and reshape to the circular shape illustrated or other nonlinear shape when deployed at a treatment site (e.g. heart chamber). The support member **54** can be made of a material having shape memory, i.e., that can be formed into a predetermined shape, straightened or bent out of its predetermined shape upon exertion of a force, and returned substantially to its predetermined shape upon removal of the force and/or application of heat. The movement and nonlinear shape of the SAS assembly **400** can be primarily determined by the properties of the support member **54**. One suitable material for the support member **54** is a nickel/titanium alloy. Such alloys typically include about 55% nickel and 45% titanium but may include from about 54% to about 57% nickel with the balance being titanium. A suitable nickel/titanium alloy is Nitinol, which has excellent shape memory, together with ductility, strength, corrosion resistance, electrical resistivity, and temperature stability.

(24) As the SAS assembly **400** moves between the linear delivery shape to the deployed nonlinear shape, the tubular member **404** can provide strain relief to the sensors **401A**, **401B**, **401C** and/or

cables **409A**, **409B**, **409C** by providing an indirect coupling to the support member **54** as opposed to a more direct coupling such as adhesion (e.g. with epoxy) to the support member **54**. The tubular member **404** can be configured to be able to slide against the support member **54** and/or skew or flex to a certain degree, essentially allowing the sensors and cables to ride as if “on rollers” over the support member **54**. In a similar manner, the tubular member **404** can act as a strain relief between a support member and sensor assembly for numerous other support member shapes and sensor configurations as understood by a person skilled in the pertinent art according to the teachings herein.

(25) Because the sensors **401A**, **401B**, **401C** are not directly adhered to the support member **54**, the sensors and cables can be manufactured over the tubular member **404** on a mandrel. The tubular member **404** with sensors and cables thereon can then be shipped as a separate component instead of being built directly on the looped support member **54**. The tubular member material can include a polymer such as polyamide, polyether ether ketone (PEEK), polyimide, or material with similar properties. The tubular member **404** can be adhered to the support member **54** at certain locations (e.g. ends) so that the tubular member **404** has some ability to slide against the support member **54** in non-affixed locations. Additionally, or alternatively, the tubular member **404** can be configured to skew or flex so that a surface of the tubular member **404** adjacent the support member **54** is capable of shifting in relation to an opposite surface to which the sensors and cables are affixed. Further, portions of the cables **409A**, **409B**, **409C** need not be adhered to the tubular member **404** so that those portions are capable of sliding to some degree in relation to the tubular member **404**, thereby providing further strain relief.

(26) FIG. 3 illustrates the distal sensor **401A** in greater detail. The distal sensor **401A** includes an inductive coil **403A** including a wire that is wound repeatedly around a predetermined length of the tubular member **404** which surrounds the support member **54**. A distal lead **406A** to the inductive coil **403A**, that is a section of the wire which forms the inductive coil **403A**, extends proximally under the coil **403A** as illustrated. Alternatively, the distal lead **406A** can extend over the coil **403A**. The distal lead **406A** and a proximal lead **407A** of the wire both extend proximally from the coil **403A** and are each joined, e.g., by wrapping and/or soldering, to a respective exposed distal end of a respective individual cable **408A** of a side-by-side dual cable **409A** at a joint region **410A** located just proximal of the coil **403A**. Alternatively, the distal lead **406A** can be joined to an individual cable of the dual cable **408A** distal from the coil **403A** and the individual cable can cross the coil **403A**. The tubular member **404** provides strain relief for the joint region **410A**, coil **403A**, and dual cable **409A**. Optionally, the junction region **410** can include additional strain relief features, such as described in U.S. Pat. Nos. 8,792,962, 10,405,774, and U.S. Patent Publication No. 2020/0015703. (See FIGS. 31-33 of the aforementioned references in particular.) In some examples, the tubular member **404** can provide sufficient strain relief at the joint region **410A** so that optional additional strain relief features can be simplified and/or eliminated.

(27) The dual cable **409A** can be wound generally transverse, for example, at least two consecutive 720° turns **421A**, around the tubular member **404** to anchor the joint region **410A** between the coil leads **406A**, **407A** and the individual cables **408A**, **408B**, thereby providing additional strain relief to the joint region **410A**. Alternatively, these generally traverse dual cable turns **421A** can be omitted in examples where the tubular member **404** provides sufficient strain relief.

(28) A protective tube **416**, for example, of polyimide, heat shrink, and/or similar material, of sufficient length can be placed over the inductive coil **403A** and joint region **410A**. Epoxy, UV glue and/or similar material filler **417A** can be injected into the tube **416A** to fill the space between the tube **416A** and the components of the sensor **401A**, with excess filler **417A** extending distally and proximally of the tube **416A** to form end caps **419A** on either side of the tube **416A**. The proximal end cap **419A** can cover at least a portion of the strain relief 720° windings **421A** of the dual cable **409A**. The filler **417A** can further provide support to the distal sensor **401A** by potting and fixing the coil **403A** and joint region **410A** onto the tubing **404** and in the tube **416A**. The filler **417A** can

provide an added degree of rigidity to the distal sensor **401A** as further protection against breakage and detachment of wire of the coil **403A** and the dual cable **409A**.

(29) The dual cable **409A** can be wrapped with looser (e.g. diagonal) windings **422A** as the dual cable **409A** extends proximally from the distal sensor **401A** along the tubular member **404**.

Optionally, additional 720° turns **420A** of the dual cable **409A** can be positioned in the proximal direction from the proximal end cap **419A** to provide additional strain relief. These additional turns **420A** are not necessary when the tubular member **404** provides sufficient strain relief.

(30) FIGS. **4** and **5** illustrate the intermediate sensor **401B** and the distal sensor **401C** respectively, which are formed in a similar manner as the proximal sensor **401A** illustrated in FIG. **3**. The intermediate sensor **401B** and the distal sensor **401C** respectively include an inductive coil **403B**, **403C** having leads **406B**, **407B**, **406C**, **407C** connected at a joint region **410B**, **410C** to individual cables **408B**, **408C** of a dual cable **409B**, **409C** covered by a tube **416B**, **416C** injected with epoxy or other material filler **417B**, **417C**. Dual wire cables **409A**, **409B** from the more distal single axis sensors **401A**, **401B** extend under the inductive coil **403B**, **403C** of the more proximal single axis sensors **401B**, **401C** as illustrated. Additionally, or alternatively, the dual wire cables **409A**, **409B** can extend over the inductive coil **403B**, **403C**. The portions of dual wire cable **409A**, **409B** which cross the inductive coils **403B**, **403C** are electrically insulated from the inductive coils **403B**, **403C** by insulation of the respective dual cable **409A**, **409B**. Similarly, each distal lead **406A**, **406B**, **406C** is electrically insulated from its respective inductive coil **403A**, **403B**, **403C** by insulation around the respective distal lead **406A**, **406B**, **406C**. Electrical insulation can be achieved by several means as understood by a person skilled in the pertinent art, and each sensor **401A**, **401B**, **401C** can be modified to include alternative electrical insulation.

(31) Referring to FIG. **4**, the dual cable **409A** from the distal sensor **401A** can include one or more traverse turns **421A** positioned near the inductive coil **403B** of the intermediate sensor **401B** distal and/or proximal of the intermediate inductive coil **403B** to provide strain relief to this dual cable **409A**. The dual cable **409B** from the intermediate sensor **401B** can include one or more traverse turns **421B** near the inductive coil **403B** and proximal of the intermediate coil **403B**. The traverse turns **421A**, **421B** near the intermediate coil **403B** can be covered by filler **417B** similar to as described in relation to FIG. **3**. In the proximal direction from the intermediate coil **403B**, the dual cable **409A** from the distal sensor **401A** and the dual cable **409B** from the intermediate sensor **401B** can be wrapped adjacent to each other. The dual cables **409A**, **409B** can optionally include additional traverse turns **420A**, **420B** in the distal and/or proximal direction from the cover of the filler **417B**. The dual cables **409A**, **409B** can include diagonal turns **422A**, **422B** to travel in the proximal direction along the tubular member **404**.

(32) FIG. **5** illustrates the dual cables **409A**, **409B**, **409C** including traverse windings **420A**, **420B**, **420C**, **421A**, **421B**, **421C** and diagonal windings **422A**, **422B**, **422C** following a similar pattern as illustrated in, and described in relation to FIGS. **3** and **4**. The dual cables **409A**, **409B**, **409C** can be jointly wound transversely and diagonally as desired proximally toward the elbow **37** (FIG. **2**).

(33) Extending over all three single axis sensors between a location immediately distal of the distal sensor **401A** and proximal of the elbow **37** but distal of the proximal end of the support member **54** is an outer nonconductive heat shrink tubular member **430**.

(34) In manufacturing the nonlinear SAS assembly **400**, the distal sensor **401A** can be formed over the tubular member **404**, followed by the intermediate sensor **401B**, and then the proximal sensor **401C**. The tubular member **404** can then be slid onto the support member **54**. The outer heat shrink tubular member **430** can be affixed over the sensors **401A**, **401B**, **401C** after the sensors are formed over the tubular member **404** and before or after the tubular member **404** is slid over the support member **54**. The SAS assembly **400** and support member **54** can be sufficiently flexible to allow movement from a linear delivery shape to a nonlinear deployed shape such as the circular region **39** illustrated in FIG. **1**.

(35) FIG. **6** is a top plan view illustration an example catheter **10**. The catheter **10** includes the

distal end section **15** which is further constructed to include a mapping assembly **27** over the SAS assembly **400**, an intermediate deflectable section **14** extending proximally from the distal end section **15**, a catheter body **12** extending proximally from the intermediate deflectable section **14**, and a control handle **16** at a proximal end of the catheter body **12** which can be used to manipulate the intermediate section **14** and/or distal end section **15** via a deflection arm **75** and/or tension adjustment dial **101**. The control handle **16** can be used to manipulate the catheter **10** similar to as described in U.S. Pat. Nos. 8,792,962, 10,405,774, and U.S. Patent Publication No. 2020/0015703 incorporated herein or other such control handle as known to a person skilled in the pertinent art.

(36) FIG. **7** is a side view illustration of the distal end section **15** including the mapping assembly **27** and intermediate section **14**. The mapping assembly **27** includes electrodes **26** configured to apply and/or receive electrical signals from tissue at a treatment site. The ring electrodes **26** can be made of any suitable solid conductive material, such as platinum or gold, or a combination of platinum and iridium, and mounted onto a non-conductive cover **52** with glue or the like. Alternatively, the ring electrodes **26** can be formed by coating the non-conductive cover **52** with an electrically conducting material, like platinum, gold and/or iridium. The coating can be applied using sputtering, ion beam deposition or an equivalent technique. A suitable mapping assembly is described in U.S. Pat. No. 7,274,957 incorporated herein by reference and attached in the Appendix of U.S. Provisional Patent Application No. 63/181,641 to which the present application claims priority. If desired, additional electrodes (not shown) can be mounted along the intermediate section **14** and/or a generally straight region **38** of the distal end section **15**.

(37) The straight region **38** is mounted on the intermediate section **14** so that it is generally a linear extension of the intermediate section **14**. The straight region **38** can have an exposed length, e.g., not contained within the intermediate section **14**, ranging from about 3 mm to about 12 mm, more preferably about 3 mm to about 8 mm, still more preferably about 5 mm, but can vary as desired. The elbow **37** is formed between the straight region **38** and the generally circular main region **39** to accommodate the angular transition therebetween.

(38) The generally circular main region **39** is generally traverse, if not also perpendicular, to the catheter body **12**. The generally circular main region **39** can form a flat circle or can be helical. The circular main region **39** can have an outer diameter ranging from about 10 mm to about 25 mm, more preferably about 12 mm to about 20 mm, and more preferably about 16 mm. The circular main region **39** can have a circumference ranging from about 30 mm to about 80 mm, more preferably about 38 mm to about 63 mm, and more preferably about 50 mm. The generally circular main region **39** can curve in a clockwise direction or a counterclockwise direction.

(39) FIG. **8** is a cross sectional illustration of the catheter **10** through the straight region **38** of the distal end section **15** as indicated by line **8-8** in FIG. **7**. Lead wires **40** to the electrodes **26** of the mapping assembly **27**, dual cables **409A**, **409B**, **409C**, the elongated support member **54**, and a contraction wire **35** can extend through the non-conductive outer tubing **52**. The contraction wire **35** can be manipulated by the handle **16** to resize the circular region **39**. The tubular member **404** of the SAS assembly **400** can also extend into the outer tubing **52**. The elongated support member **54** and the contraction wire **35** can be positioned within the tubular member **404**. The dual cables **409A**, **409B**, **409C** of the SAS assembly **400** can be contained within a protective, non-conductive tube **413**.

(40) The outer tubing **52** can have any suitable cross-sectional shape as desired. The outer tubing **52** can be made of any suitable material and is preferably made of a biocompatible plastic such as polyurethane or PEBAX. The outer tubing **52** can be pre-formed into the desired generally circular shape of the generally circular main region **39**. Additionally, or alternatively, the shape of the generally circular main region **39** can be defined by a wire or other component extending through the outer tubing **52** such as the support member **54** of the SAS assembly **400**.

(41) FIG. **9** is a cross sectional illustration of the catheter **10** through the intermediate section **14** as indicated by line **9-9** in FIG. **7**. The intermediate section **14** includes four lumens **30**, **31**, **32**, **33**.

The intermediate section **14** can be otherwise configured with more or fewer lumens as understood by a person skilled in the pertinent art. The dual cables **409A**, **409B**, and **409C** of the SAS assembly **400** and the lead wires **40** to the electrodes **26** can traverse a first lumen **30** of the four intermediate section lumens. As illustrated, two deflection puller members **42** each respectively extend through opposite lumens **31**, **33** of the intermediate section **14** and are each respectively surrounded by a compression coil **44**. The deflection puller members **42** can be manipulated by the handle **16** to deflect the intermediate section **14** as illustrated in FIG. **6**. A fourth lumen **32** of the intermediate section **14** includes the contraction wire **35** surrounded by another compression coil **46**.

(42) FIG. **10** is a cross sectional illustration near a distal dome **51** of the distal end section **15**. As illustrated, the distal end section **15** is sealed closed with the dome **51**. The dome **51** can include polyurethane glue or the like. A short ring **56**, made of metal or plastic (e.g., polyamide) is mounted at the distal end of the non-conductive outer tubing **52**. The short ring **56** prevents the distal end of the outer tubing **52** from collapsing, thereby maintaining the diameter of the outer tubing **52** at its distal end. Material of the dome can be attached to a length of about 1 mm to about 2 mm of an exterior surface of the outer tubing **52** to anchor the dome **51** to the outer tubing **52**.

(43) The support member **54** and the distal end of the contraction wire **35** are soldered or otherwise attached to a small stainless steel tube **53** positioned within the outer tubing **52**. With this arrangement, the relative positions of the contraction wire **35** and the support member **54** can be controlled so that the contraction wire can be positioned on the side of the generally circular region **39** closer to the center of the generally circular region **39**. The contraction wire **35** on the inside of the curve pulls the support member **54** to the inside of the curve, enhancing contraction of the generally circular region **39**. An inner tubing **404** surrounding the support member **54** and contraction wire **35** is preferably the same tubing as the tubular member **404** of the SAS assembly **400**, although the inner tubing can be separate from the tubular member **404**. The inner tubing **404** can include a braided layer to inhibit the contraction wire **35** from tearing through the inner tubing **404**. The outer tubing **52** can include openings **57** through which the lead wires **40** can pass to make electrical contact with the electrodes **26**.

(44) FIG. **11** is a cross sectional illustration of a junction between the intermediate section **14** and the catheter body **12** as indicated in FIG. **7**. The illustrated cross section is taken at a first diameter that is through the first lumen **30** and fourth lumen **32** as oriented vertically in relation to FIG. **9**.

(45) FIG. **12** is a cross sectional illustration of the junction illustrated in FIG. **11** as indicated in FIG. **7**. The illustrated cross section is taken at a second diameter through lumens **31**, **33** respectively having a contraction wire **42** extending therethrough, where the second diameter is orthogonal to the first diameter and aligned horizontally in relation to FIG. **9**.

(46) Referring collectively to FIGS. **11**, and **12**, the catheter body **12** can include a single, central or axial lumen **18**. The catheter body **12** can be flexible, i.e., bendable, and substantially non-compressible along its length. The catheter body **12** can be of any suitable construction and made of any suitable material. A suitable construction can include an outer wall **22** made of a polyurethane or nylon. The outer wall **22** can include an imbedded braided mesh of stainless steel or the like to increase torsional stiffness of the catheter body **12** so that, when the control handle **16** is rotated, the distal end section **15** of the catheter **10** will rotate in a corresponding manner. A single lumen catheter body **12** may be preferred over a multi-lumen body because the single lumen body can permit better tip control when rotating the catheter **10**. The single lumen **18** permits the components passing therethrough to float freely within the catheter body **12**. If such components were restricted within multiple lumens, they can build up energy when the handle **16** is rotated, resulting in the catheter body **12** having a tendency to rotate back if, for example, the handle is released, or if bent around a curve, to flip over, either for which are undesirable performance characteristics.

(47) The outer diameter of the catheter body **12** is not critical but is preferably no more than about

8 French. Likewise, the thickness of the outer wall **22** is not critical. The inner surface of the outer wall **22** can be lined with a stiffening tube **20**, which can be made of any suitable material, e.g., polyimide. The stiffening tube **20** can be held in place relative to the outer wall **22** at the proximal end of the catheter body **12**. A first glue joint **23** can be made between the distal ends of the stiffening tube **20** and the outer wall **22** by a fast-drying glue, e.g. Super Glue®. Thereafter a second glue joint **25** can be formed between the proximal ends of the stiffening tube **20** and outer wall **22** using a slower drying but stronger glue, e.g., polyurethane.

(48) The stiffening tube, along with the braided outer wall **22**, can provide torsional stability. The outer diameter of the stiffening tube **20** can be about the same as or slightly smaller than the inner diameter of the outer wall **22**. Polyimide tubing is suitable because it may be very thin walled while still providing very good stiffness. This maximizes the diameter of the central lumen **18** without sacrificing strength and stiffness. Polyimide material is typically not used for stiffening tubes because of its tendency to kink when bent. However, in combination with an outer wall **22** of polyurethane, nylon or other similar material, particularly having a stainless steel braided mesh, the tendency for the polyimide stiffening tube **20** to kink when bent is essentially eliminated with respect to the applications for which the catheter is used.

(49) The outer wall **22** can have an outer diameter of about 0.092 inch (about 2.3 mm) and an inner diameter of about 0.063 inch (about 1.6 mm) and the polyimide stiffening tube **20** has an outer diameter of about 0.029 inch (about 1.56 mm) and an inner diameter of about 0.052 inch (about 1.3 mm).

(50) Referring collectively to FIGS. **9**, **11**, and **12**, the intermediate section **14** can include a shorter section of tubing **19** with multiple off-axis lumens, for example, first, second, third and fourth lumens **30**, **31**, **32** and **33**. The tubing **19** can be made of a suitable non-toxic material which is preferably more flexible than the catheter body **12**. A suitable material for the tubing **19** can be braided polyurethane, i.e., polyurethane with an embedded mesh of braided stainless steel or the like. The outer diameter of the intermediate section **14**, like that of the catheter body **12**, is preferably no greater than about 8 French. The size of the lumens is not critical. In one embodiment, the intermediate section has an outer diameter of about 7 French (0.092 inch, 2.3 mm) and the lumens are generally about the same size, having a diameter of about 0.022 inch (about 0.56 mm), or selected lumens can have a slightly larger diameter of about 0.036 inch (about 0.91 mm).

(51) The proximal end of the intermediate section **14** can include an inner counter bore **24** that receives the outer surface of the polyimide stiffener **20**. The intermediate section **14** and catheter body **12** can be attached by glue **29** or the like.

(52) Various components, such as lead wires **40**, dual cables **409A**, **409B**, **409C**, and contraction wires **35**, **42** can extend through the single lumen **18** of the catheter body **12**. Longitudinal movement of the contraction wires **35**, **42** to the catheter body **12** enable user control of various parts of the catheter via the control handle **16**. Distal ends of the deflection puller members **42** can be anchored to the wall of the tubing **19** near the distal end of the intermediate section **14**. In the intermediate section **14**, each of the deflection puller members **42** extends through a plastic, e.g., Teflon®, tubular member **81**, which prevents the deflection puller members **42** from cutting into the wall of the tubing **19** of the intermediate section **14** when the intermediate section **14** is deflected.

(53) Compression coils **44** can surround the deflection puller members **42** and can extend from the proximal end of the catheter body **12** to the proximal end of the intermediate section **14**. The compression coils **44** can be made of any suitable metal, e.g., stainless steel. The compression coils **44** can each be tightly wound on itself to provide flexibility, i.e., bending, but to resist compression. The inner diameter of the compression coils **44** is preferably slightly larger than the diameter of the puller wires **42**. For example, when a puller member **42** has a diameter of about 0.007 inches (about 0.18 mm), the corresponding compression coil **44** preferably has an inner diameter of about 0.008

inches (about 0.20 mm). The Teflon® coating on the puller member **42** allows each puller member **42** to slide freely within the corresponding compression coil. The outer surface of the compression coils can be covered by a flexible, non-conductive tubular member to prevent contact between the compression coils and other components, such as lead wires and cables, etc. A non-conductive tubular member can be made of polyimide tubing or other suitable material.

(54) The compression coils **44** can be anchored at their proximal ends to the proximal end of the stiffening tube **20** in the catheter body **12** by a glue joint **50** and at its distal end near the proximal end of the intermediate section **14** in the second lumen **31** and fourth lumen **33** by glue joints **51**.

(55) A third compression coil **46** can be situated within the catheter body **12** and intermediate section shaft **14** surrounding the contraction wire **35** (FIG. **8A**). The third compression coil **46** extends from the proximal end of the catheter body **12** to near the distal end of the third lumen **32** of the intermediate section **14**. The third compression coil **46** can be made of any suitable metal, e.g., stainless steel, and is tightly wound on itself to provide flexibility, i.e., bending, but to resist compression. The inner diameter of the third compression coil **46** is preferably slightly larger than the diameter of the contraction wire **35**. The outer surface of the compression coil **46** can be covered by a flexible, non-conductive tubular member **68**, e.g., made of polyimide tubing. The third compression coil **46** can be formed of a wire having a square or rectangular cross-sectional area, which makes it less compressible than a compression coil formed from a wire having a circular cross-sectional area. As a result, the third compression coil **46** can be configured to keep the catheter body **12**, and particularly the intermediate section **14**, from deflecting when the contraction wire **35** is manipulated to contract the mapping assembly **17** as it absorbs more of the compression.

(56) The third compression coil **46** can be anchored at its proximal end to the outer wall **22** of the catheter body **12** by the proximal glue joint **50** and to the intermediate section **14** by distal glue joint **72**.

(57) The lead wires **40** attached to the ring electrodes **26** extend through the first lumen **30** of the intermediate section **14**, through the central lumen **18** of the catheter body **12**, through the control handle **16**, and terminate at their proximal end in a connector (not shown) which is connected to an appropriate monitor or other device for receiving and displaying the information received from the ring electrodes **26**. The portion of the lead wires **40** extending through the central lumen **18** of the catheter body **12**, control handle **16** and proximal end of the intermediate section **14** is enclosed within a protective tubular member **62**, which can be made of any suitable material, such as polyimide. The protective tubular member **62** is anchored at its distal end to the proximal end of the intermediate section **14** by gluing it in the lead wire lumen **30** with polyurethane glue or the like to form glue joint **73**.

(58) The lead wires **40** are attached to the ring electrode **26** by any conventional technique. In one embodiment, each ring electrode **26** is mounted by first forming a hole in the non-conductive cover **52**. An electrode lead wire **40** is fed through the hole, and the ring electrode **26** is welded in place over the lead wire and non-conductive cover **52**.

(59) FIGS. **13A** through **13E** illustrate steps of constructing a distal end of the example catheter **10**. FIG. **13A** illustrates the SAS assembly **400** over a mandrel **500**. The tubular member **404** is placed over the mandrel **500**. The sensors **401A**, **401B**, **401C** and the cables **409A**, **409B**, **409C** can be mounted to the tubular member **404** while the tubular member **404** is over the mandrel **500**. FIG. **13A** includes a cut-out for the purposes of illustration showing a cable **409A** wound over the tubular member **404**. The mandrel is illustrated having a substantially linear geometry. Alternatively, the mandrel can have a non-linear geometry to facilitate movement of the SAS assembly **404** from the linear configuration to the nonlinear shape when the distal portion **15** of the catheter **10** is deployed.

(60) FIG. **13B** is a cross-sectional view of the mid encapsulated SAS **401B** as indicated in FIG. **13A**. The sensors **401A**, **401B**, **401C** and the cables **409A**, **409B**, **409C** can be secured to the

tubular member **404** by a protective tube **416A**, **416B**, **416C** (for example, of polyimide, heat shrink, and/or similar material) and an adhesive **417A**, **417B**, **417C** (for example epoxy, UV glue and/or similar material). The protective tubes **416A**, **416B**, **416C** can extend only over the coils **403A**, **403B**, **403C** such as illustrated in FIGS. 3 through 5. Optionally, the SAS assembly **400** need not include the protective tubes **416A**, **416B**, **416C**, in which case heat shrink **430** or other tubing extending over the length of the SAS assembly **400** can function to protect the sensors **401A**, **401B**, **401C**.

(61) FIG. 13C illustrates the SAS assembly **400** removed from the mandrel **500**. Once the SAS assembly **400** is constructed, it can be shipped as a component that can be assembled with the remainder of the catheter **10**.

(62) FIG. 13D illustrates the SAS assembly **400** placed onto a support member **54** in a circular shape.

(63) FIG. 13E is a cross-sectional view of the mid encapsulated SAS **401B** as indicated in FIG. 13D.

(64) The distal section **15** of the example catheter **10** illustrated herein can be modified in one or more ways in an effort to reduce breakage of cables **409A**, **409B**, **409C**. In some examples, each dual cable **409A**, **409B**, **409C** can be wound generally transverse for an increased number of consecutive 720° turns **421A**, **421B**, **421C** around the tubular member **404** or around the support member **54**. The increased number of traverse turns **421A**, **421B**, **421C** is preferably about 5 to about 9 turns **421A**, **421B**, **421C** and more preferably about 5 to about 7 turns. The number of traverse turns **421A**, **421B**, **421C** can be increased at least on a distal side of each coil **403A**, **403B**, **403C**. The increased number of traverse turns **421A**, **421B**, **421C** can act as reinforcement if epoxy wicks onto the cables **409A**, **409B**, **409C**. The increased number of traverse turns **421A**, **421B**, **421C** can increase thickness of the wire along an axis of high strain. In some examples, a shrink sleeve can be added over the traverse turns **421A**, **421B**, **421C**, and/or the protective tube **416A**, **416B**, **416C** can be extended to completely cover the traverse turns **421A**, **421B**, **421C**. The shrink sleeve and/or extended protective tube **416A**, **416B**, **416C** can mitigate epoxy from wicking onto the cables **409A**, **409B**, **409C**. A transition point **41** (FIG. 16) for wire shield can be positioned on the straight region **38**, **738** of the distal end section **15** rather than in the circular region **39** or near the elbow **737**, **37** (FIG. 2). By positioning the transition point **41** away from the sharp radii of the elbow **737**, **37**, cables **409A**, **409B**, **409C** may be less likely to break at the transition point **41**. The cables **409A**, **409B**, **409C** are preferably about 50 AWG.

(65) FIG. 14 is a side view of an intermediate sensor **503B** of a variation of the example catheter **10**. The intermediate sensor **503B** is wound about a support member **554**. The intermediate sensor **503B** and support member **554** can be constructed similar to corresponding components **403B**, **54** disclosed elsewhere herein. The illustrated segment includes a first dual cable **508A** electrically connected to a distal sensor (not shown, configured similar to distal sensor **403A**) positioned in a distal direction (DD) in relation to the intermediate sensor **503B**. The first dual cable **508A** winds about the support member **554** from the distal sensor in a proximal direction (PD) toward the intermediate sensor **503B**. Near a distal side of the intermediate sensor **503B**, the first dual cable **508A** makes seven (7) consecutive traverse turns **521A** of approximately 720° about the support member **554**. The increased number of traverse turns **521A** can reduce the likelihood of cable breakage at the distal side of the intermediate sensor **503B** compared to a construction that is identical but for having a reduced number of traverse turns **521A** on the distal side of the intermediate sensor **503B**. The support member **554** defines a longitudinal axis L-L of the distal section **15** of the catheter **10** near the intermediate sensor **503B**. As the support member **554** moves between a circular configuration and a straight configuration, the increased number of traverse turns **521A** on the distal side of the intermediate sensor **503B** can provide strain relief for the first dual cable **508A** and inhibit range of motion of the support member **554** at the distal side of the intermediate sensor **503B**. The increase number of traverse turns can force the wires to align more

or less perpendicular to the longitudinal axis L-L of the support member **554**. Given that the tensile loads induced by straightening the support member **554** are parallel to the longitudinal axis L-L, having the dual cables perpendicular to this direction can reduce the stress and strain on the cross section of the dual cables.

(66) As illustrated in FIG. **14**, the first dual cable **508A** can travel over an exterior of the intermediate sensor **503B**. Additionally, or alternatively, the first dual cable **508A** can travel under the intermediate sensor **503B**. A second dual cable **508B** can be electrically connected to the intermediate sensor **503B**, include one or more traverse turns **531B** on the proximal side of the intermediate sensor **503B**, and wind in the proximal direction (PD) about the support member **554**. Although not shown, the first and/or second dual cable **508A**, **508B** can include an increased number of traverse turns on the proximal side of the intermediate sensor **503B**. The dual cables **508A**, **508B** can wrap with diagonal windings **522A**, **522B** between sensors.

(67) The first dual cable **508A** and/or second dual cable **508B** can similarly be wrapped with an increased number of consecutive traverse turns (compared to FIGS. **1-5**) on a distal side of a proximal sensor (not shown, configured similar to proximal sensor **403C**) that is in the proximal direction (PD) in relation to the intermediate sensor **503B**. The SAS assembly can further include a third dual cable (not shown, configured similar to third dual cable **408C**). The first, second, and/or third dual cables can be wrapped with an increased number of consecutive traverse turns on a proximal side of the proximal sensor.

(68) FIG. **15** is a side view of an intermediate sensor **603B** of another variation of the example catheter **10** showing an increased number of traverse turns **621A** (compared to FIG. **14**) of a first dual cable **608A** positioned on a distal side of the intermediate sensor **603B**. The increased number of turns can provide strain relief for the first dual cable **608A** and inhibit range of motion of the support member **554** at the distal side of the intermediate sensor **603B**. An epoxy, UV glue, and/or similar material filler **617B** can be applied over the intermediate sensor **603B** and adjacent windings of dual cables (e.g. at least over traverse windings **621A** of the first dual cable **608A** as illustrated). A protective tube **616B**, for example, of polyimide, heat shrink, and/or similar material, of sufficient length can cover the inductive coil **603B** adjacent windings, and filler **617B**. In some processes, the filler **617B** can be injected into the tube **616B** similar as described in relation to FIG. **3** with an exception that the filler **617B** is completely covered by the protective tube **616B**, i.e. the filler **617B** does not form end caps **419A** on either side of the tube **416A** as illustrated in FIG. **3**. The filler **617A** and protective tube **616B** can provide an added degree of rigidity to the intermediate sensor **603B** as further protection against breakage of the dual cable **608A**. The catheter **10** can be similarly modified at the distal sensor **401A** and the proximal sensor **401C**. The filler **617B** and protective tube **616B** can be similarly configured in variations of the catheter **10** having fewer traverse windings such as in FIGS. **3-5** and FIG. **14**.

(69) FIG. **16** is a side view illustration of a variation of the example catheter **10** showing an alternative distal section **715** configuration. Certain components are removed for the sake of illustration. The distal section **715** is constructed similarly to as illustrated in FIG. **2** with an exception that the transition point **41** is positioned on a straight region **738** away from the elbow **37** so that cables **409A**, **409B**, **409C** may be less likely to break at the transition point **41**. The distal section **715** includes dual cables **709A**, **709B**, **709C** configured similarly to dual cables disclosed elsewhere herein. The degree of bend of the elbow **737** can be defined by an elbow diameter (ED) as illustrated. The elbow **737** is positioned between a straight region **738** and a circular region **739**. According to aspects of the present disclosure, the elbow diameter (ED) can be sized to reduce likelihood of cable break due to flexion of the distal section **715** at the elbow **737**. The elbow diameter is preferably greater than 0.18 inches, and more preferably the elbow diameter is about 0.26 to about 0.23 inches in diameter. Strain induced when the distal section **715** moves between a straight configuration and a circular configuration is less for a larger elbow diameter (ED).

(70) The descriptions contained herein are illustrative examples and are not intended in any way to

limit the scope of the claimed invention. Examples can be modified, e.g. by utilizing alternative materials and alternative geometries of component parts. Modifications apparent to those skilled in the pertinent art to which this disclosure relates and are intended to be within the scope of the claims which follow.

Claims

1. A single axis sensor (SAS) assembly comprising: a tubular member comprising a lumen therethrough sized to receive an elongated support member suitable for shaping a distal section of a mapping catheter; a distal inductive coil affixed to the tubular member and coaxial to the tubular member and comprising leads; a first dual cable electrically joined to the leads of the distal inductive coil and wound over the tubular member and an elbow of the distal section of the mapping catheter; a proximal inductive coil affixed to the tubular member across the first dual cable, coaxial to the tubular member, and separated from the distal inductive coil such that the distal inductive coil moves in relation to the proximal inductive coil when the tubular member is moved from a linear configuration to a nonlinear configuration; and an intermediate inductive coil affixed to the tubular member, coaxial to the tubular member, positioned between the distal inductive coil and the proximal inductive coil when the SAS assembly is in the linear configuration, the first dual cable being wound over a distal end of a first protective tube, the first protective tube covering the intermediate inductive coil.
2. The SAS assembly of claim 1, the nonlinear configuration being approximately circular with a circumference measuring from 24 millimeters 96 millimeters.
3. The SAS assembly of claim 1, the lumen comprising a diameter measuring from 0.08 mm to 0.3 mm.
4. The SAS assembly of claim 1, further comprising: a second dual cable electrically joined to leads of the intermediate inductive coil and wound over the tubular member to cross the proximal inductive coil.
5. The SAS assembly of claim 1, the nonlinear configuration being circular, and the proximal inductive coil, intermediate inductive coil, and distal inductive coil being positioned equidistant around a circumference of the nonlinear configuration.
6. The SAS assembly of claim 1, the first dual cable being wound for 5 to 7 consecutive traverse turns of 720° around the first protective tube covering the intermediate inductive coil and the tubular member adjacent the distal end of the intermediate inductive coil.
7. The SAS assembly of claim 6, further comprising: a second protective tube positioned to completely cover the traverse turns.
8. The SAS assembly of claim 1, the elbow comprising a diameter of 0.18 inches to 0.26 inches.
9. The SAS assembly of claim 1, further comprising a shrink sleeve extending over the length of the SAS assembly.
10. The SAS assembly of claim 8, further comprising: a shield wire transition point positioned on a straight region of the elongated support member and away from the elbow, the straight region maintaining a straight configuration when the elongated support member moves to the nonlinear configuration.
11. A mapping catheter comprising: an elongated support member extending through a distal section of the mapping catheter and comprising a nonlinear predetermined configuration to which the elongated support member moves when the distal section is disposed within a patient; and a single axis sensor (SAS) assembly comprising: a tubular member surrounding the elongated support member, a distal inductive coil affixed to the tubular member and coaxial to the tubular member, a proximal inductive coil affixed to the tubular member, coaxial to the tubular member, and separated from the distal inductive coil such that the distal inductive coil moves in relation to the proximal inductive coil when the elongated support member moves to the nonlinear

predetermined configuration, a first dual cable electrically joined to leads of the distal inductive coil and wound over the tubular member to cross the proximal inductive coil, the first dual cable further being wound about an elbow of the distal section of the mapping catheter; and an intermediate inductive coil affixed to the tubular member, coaxial to the tubular member, positioned between the distal inductive coil and the proximal inductive coil when the elongated support member is linearly elongated, the first dual cable being wound over a distal end of a first protective tube, the first protective tube covering the intermediate inductive coil.

12. The mapping catheter of claim 11, the intermediate inductive coil positioned such that the proximal inductive coil, intermediate inductive coil, and distal inductive coil collectively function as a three-axis sensor when the elongated support member moves to the nonlinear predetermined configuration.

13. The mapping catheter of claim 12, the nonlinear predetermined configuration being circular, and the proximal inductive coil, intermediate inductive coil, and distal inductive coil being positioned equidistant around a circumference of the nonlinear predetermined configuration.

14. The mapping catheter of claim 13, the nonlinear predetermined configuration comprising a circumference measuring from 10 millimeters to 50 millimeters.

15. The mapping catheter of claim 12, the SAS assembly further comprising: a second dual cable electrically joined to leads of the intermediate inductive coil and wound over the tubular member to cross the proximal inductive coil.

16. The mapping catheter of claim 12, the first dual cable being wound for 5 to 7 consecutive traverse turns of 720° around the first protective tube covering the intermediate inductive coil and the tubular member adjacent the distal end the tubular member adjacent a side of the respective coil.

17. The mapping catheter of claim 16, the SAS further comprising a second protective tube positioned to completely cover the traverse turns.

18. The mapping catheter of claim 11, the elongated support member comprising a memory shape material heat set to the nonlinear predetermined configuration.
